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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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1. Team & Project Overview

Team Name	#slaynergy
Project / Solution Name	Jobbery
Name (Leader)	Yang Zi Chien
Email & Phone (Leader)	zichienyang@gmail.com / 0139681431
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	http://sdc-runway.vercel.app/
Code Repository URL (public / access granted to judges)	https://github.com/yabgzichien/SDCHack-slaynergy (public, Apache-2.0 licence)

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

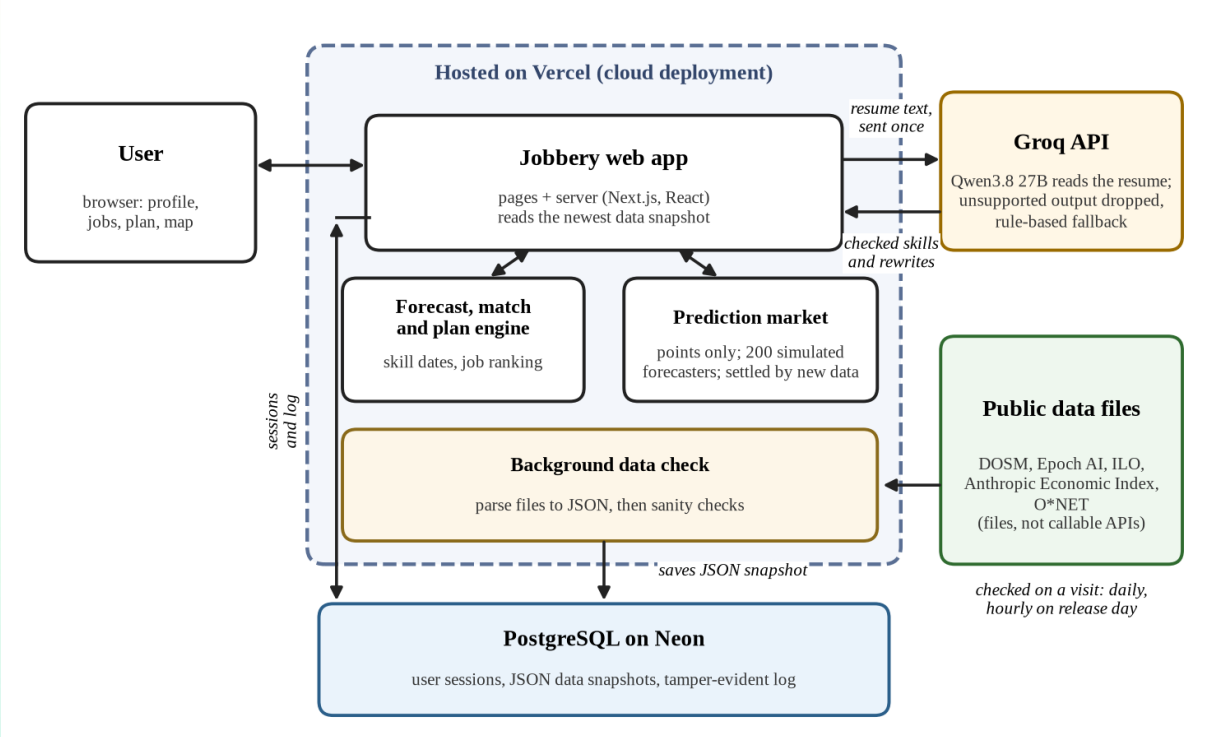
A user interacts with Jobbery in a web browser, deployed on Vercel.

The app does three things. First, it runs our own forecasting, matching and planning engine on dated public data from DOSM, Epoch AI, the ILO, the Anthropic Economic Index and O*NET. These sources are not callable APIs, so a Vercel Cron job (daily) and visitor checks (hourly on a DOSM release day) download the files and parse them into JSON snapshots in the database.

A new release is accepted only if it passes sanity checks, New will data settles our points-only prediction market (the prediction market is seeded with 200 simulated forecasters) and updates every user's skill job ranking and plan.

Next, when the user uploads a resume and leaves the AI option on, the app sends the text once to an LLM: the KrackedDev AI API first, then RunPod, then Groq. The app drops anything unsupported, blocks prompt injections, and a rule-based scanner takes over if all fail. The same LLM only helps find DOSM's release page when the plain rules fail. Lastly, the app saves the user's session in a Neon PostgreSQL database, with a tamper-evident log. See the diagram below.

Figure 1. System block diagram



2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9.3	Programming language for the whole app	Apache-2.0
Node.js 22	Runs the app on the server	MIT
Next.js 16.4.2; React 19.3.0 (with react-dom)	Web framework: pages, server and the screens	MIT
Tailwind CSS 4.3.3; Framer Motion 14.4.0	Styling; page and scroll animation	MIT
Recharts 3.10.1	Charts for trends and forecasts	MIT
Lucide React 1.60.0	Icons	ISC
pg 8.23.1; better-sqlite3 13.0.3	Connects the app to PostgreSQL; local test database on a laptop (not used in the live deployment)	MIT
unpdf 1.9.1	Reads text from PDF resumes	MIT
mammoth 1.13.0	Reads text from Word resumes	BSD-2-Clause
fflate 0.8.3	Unpacks downloaded data files when building the data snapshots	MIT
Vitest 5.0.5; ESLint 9.39.5; tsx 4.23.15	Automated tests (242), code checking and running our build scripts	MIT
Instrument Sans (font, via Google Fonts)	Text font; fetched when the app is built	SIL Open Font Licence 1.1

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
KrackedDev AI API (LiteLLM gateway), model deepseek-v4.1-flash	First choice for the optional resume reader and the course-certificate reader: suggests extra skills (each with a quote from the resume) and better wording for weak lines; the worker decides what to use. Also the data-source finder: only if the plain rules cannot find or read DOSM's release, it picks the release page from DOSM's own list and says where the table sits in a changed workbook. It never supplies a number.	Access key issued to our team (not sponsor-provided)
RunPod Serverless (RunPod, Inc.), Qwen/Qwen2.5-14B-Instruct on vLLM	Second choice: the same resume and certificate reading when the first call fails or times out.	Sponsor-provided (US$25 team credit)
Groq API (Groq, Inc.), model qwen/qwen3.8-27b	Last backup for the same resume and certificate reading. If all three fail, the rule-based scanner answers.	Free tier (not sponsor-provided)
None other	No other external API is called while the app runs. The server itself downloads public data files from DOSM, OpenDOSM and Epoch AI, and checks the ILO and Anthropic Economic Index pages for new versions. The map is drawn by us, and no payment, messaging or map service is used.	Not applicable

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Vercel	Hosts the live web app and runs the daily Vercel Cron job (/api/cron/refresh) that refreshes the public data	Free (Hobby) plan, Singapore (Southeast) - ap-southeast-1 - sin1; cron runs once a day
PostgreSQL database, Neon	Stores worker sessions (profile, plan progress), advisor bookings, the activity log and the versioned data snapshots saved by each refresh	Free plan, Asia Pacific 1 Singapore
RunPod Serverless	Runs the Qwen2.5 14B Instruct endpoint used as the second-choice AI resume and certificate reader; billed only while a worker runs	Serverless endpoint, US$25 sponsor credit
GitHub	Public source code repository (Apache-2.0)	Free plan

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
DeepSeek V4.1 Flash (deepseek-v4.1-flash)	DeepSeek; served by the KrackedDev AI API	Pre-trained, used as-is (no fine-tuning). Resume reader, certificate reader and DOSM data-source finder	The terms of the KrackedDev AI API and of the model's publisher apply
Qwen2.5 14B Instruct (Qwen/Qwen2.5-14B-Instruct)	Alibaba Qwen team; run by us on RunPod Serverless (vLLM)	Pre-trained, used as-is (no fine-tuning). Second-choice resume and certificate reader	Apache 2.0 (open weights)
Qwen3.8 27B (qwen/qwen3.8-27b)	Alibaba Qwen team; served by the Groq API (last backup reader)	Pre-trained, used as-is (no fine-tuning)	Apache 2.0 (open weights). Groq lists the model as Preview and its own terms apply.
Jobbery forecasting engine: three AI-speed futures plus a prediction market	Built by our team	Self-built statistical model	Our code, Apache-2.0
About 200 simulated forecasters	Built by our team	Rule-based simulation, always labelled as made-up	Our code, Apache-2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
DOSM Job Market Insights, 2026 Q2 (published 18 Aug 2026)	dosm.gov.my/portal-main/release-content/big-data-analytics-job-market-insights-q22026	Public	Malaysian Government Open Data Terms of Use 1.0 (credit DOSM)	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
OpenDOSM employment, unemployment and wage data by state	open.dosm.gov.my	Public	CC BY 4.0	N
Epoch AI benchmarking data (models to 15 Oct 2026)	epoch.ai/benchmarks	Public	CC BY 4.0	N
Anthropic Economic Index (2025-02-10 release)	huggingface.co/datasets/Anthropic/EconomicIndex	Public	CC BY (data)	N
ILO generative-AI exposure scores (2025 update)	ilo.org/publications/generative-ai-and-jobs-refined-global-index-occupational-exposure	Public	CC BY 4.0	N
O*NET occupation and task data	onetcenter.org/database.html	Public	CC BY 4.0	N
geoBoundaries Malaysia state borders	geoboundaries.org	Public	ODbL 1.0 (OpenStreetMap)	N
UN SDG target text	sdgs.un.org/goals	Public	UN text, quoted with source	N
Course list: 97 courses we checked by hand (title, provider, hours, link, check date)	Links to each provider’s own page	Self-collected	Links only; no course content copied	N
Demo worker profiles (4) and simulated forecasters (200)	Made by our team	Synthetic, always labelled	Our work, Apache-2.0	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑️ Yes No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic), using Claude Opus 5.5 and Claude Sonnet 5.5	Writing, testing and revising the app code and data scripts.	Major
Chatgpt Codex (OpenAI), using 6.1Sol and Terra	Front-end design	Major
Replit Core AI	Initial web app framework and design, and code writing	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

A team member read and ran every AI-written change before we commit and keep the changes, and we re-run them and make sure everything works as we intended before each deployment. For the part that uses LLM API, we make sure that prompt injection security issue is not plausible and set a guardrail for the LLM to make sure that it will never invent new changes or hallucinate. The data-fetching model never supplies numbers, and every release must pass validation checks. And also we checked the data integration and make sure that the data is integrated properly.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑️ Yes No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core AI	Use for code generation
RunPod	RunPod Serverless (US$25 credit)	Runs Qwen2.5 14B Instruct, the second-choice AI resume and certificate reader

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

The sponsor resources were not essential to the core functionality of the prototype, however, Replit Core AI was a significant part of building our project, we use it to help with code generation and software engineering. RunPod only serves as a backup for the optional resume reader: if it is unavailable, another provider or the rule-based scanner answers.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Jobbery does not create accounts, each visitor gets an anonymous session that lasts 7 days. During that session, it keeps only what the user inputs, namely a name last job, qualification, skills, interests, months and weekly hours available to study, and optionally age and home state. Age and gender are never used in matching.

Resumes are never stored in the database. However, if the AI option is on, the resume is sent to the KrackedDev AI API, or to RunPod and then Groq if that fails, where their terms apply. Uploaded certificates are handled the same way.

All of this is stored in a Neon PostgreSQL database, and the activity log holds only an anonymous ID and fingerprints. The data stays in the database until the worker presses "Start over" button, which deletes their profile and plan.

In future work we would add authentication to keep users' data secure.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Advisor booking has no identity check or no-show protection. Anyone can open Jobbery and book an advisor slot without proving who they are. Fake bookings could take slots real job seekers need, and a no-show wastes an advisor's time. Today's safeguards are small. Future work**:** require sign-in with a verified email or phone number before booking, then add a refundable commitment deposit that is forfeited on a no-show or late cancellation. We have not built this yet.

Other limits: forecasts are estimates and our 200 forecasters are simulated forecaster not a real people. Future work: we would attract real people to join in to have a better predicting edge of the real cause.. The resume and data-fetching models can be wrong or unavailable: they only propose, every figure is checked by rules, and the rule-based scanner answers if every provider fails.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑️ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑️ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑️ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑️ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑️ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑️ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑️ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team’s knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



Name: Yang Zi Chien
31 Oct 2026